Assignment 02



6 BAR TOGGLE MECHANISM

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SECTION 1

Static force analysis

A constant force of **10 N**, acting in a fixed global X- axis direction, is applied to a slider in the mechanism. Using **ADAMS software**, we aim to determine the required force or torque at the driving component to maintain equilibrium under these conditions. The analysis is conducted at four key crank positions: **45°, 135°, 225°, and 315°**.Although the applied force maintains a consistent global direction, its point of application on the output component varies intentionally. These variations lead to different force or torque requirements for stabilizing the mechanism at each crank position. To illustrate the effects of these forces, snapshots of the simulation results are provided, offering a clear visualization of the force interactions within the system.

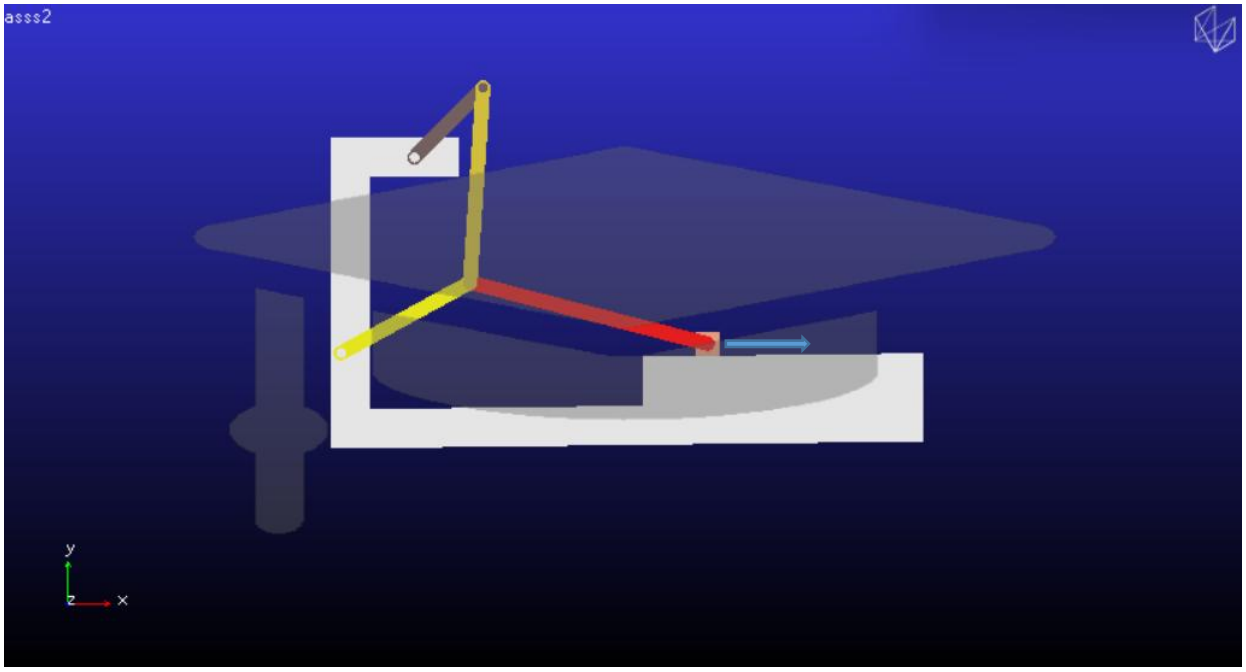


Fig 1. Mechanism at 45° position

Static force on 45°

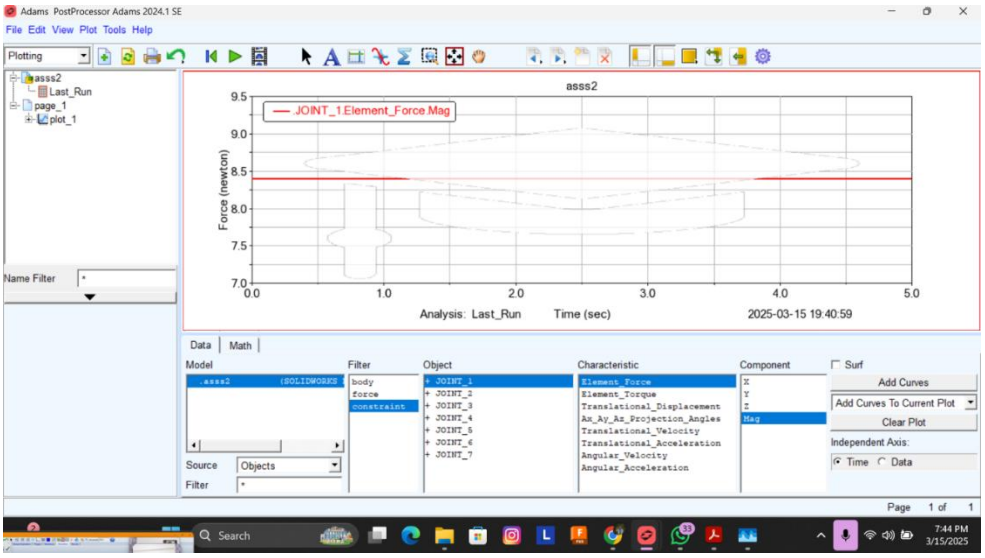


Fig 2. Force vs. Time

Static torque on 45°

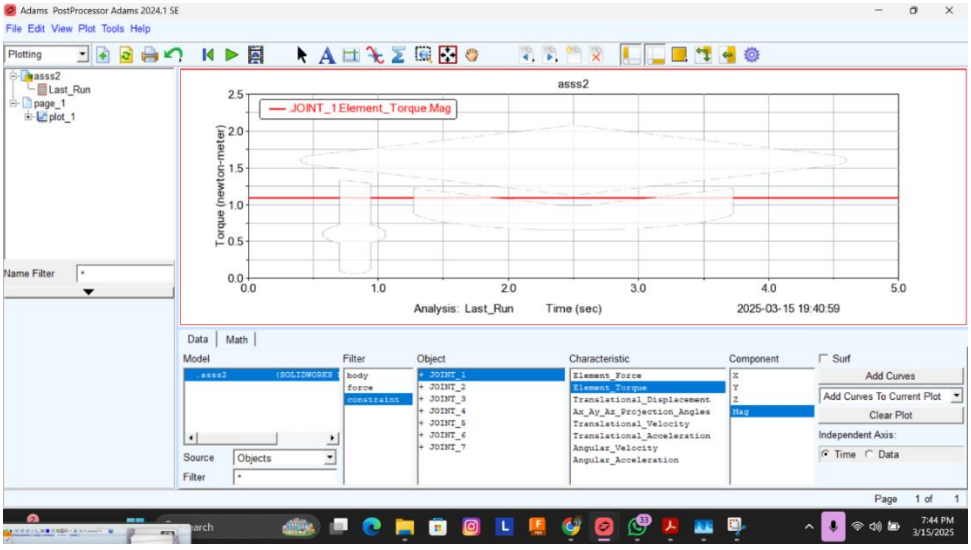


Fig 3. Torque vs. Time

From above figure it is seen that force 8.4N and torque 1.15Nm in the crank is required to hold the mechanism in this equilibrium position.

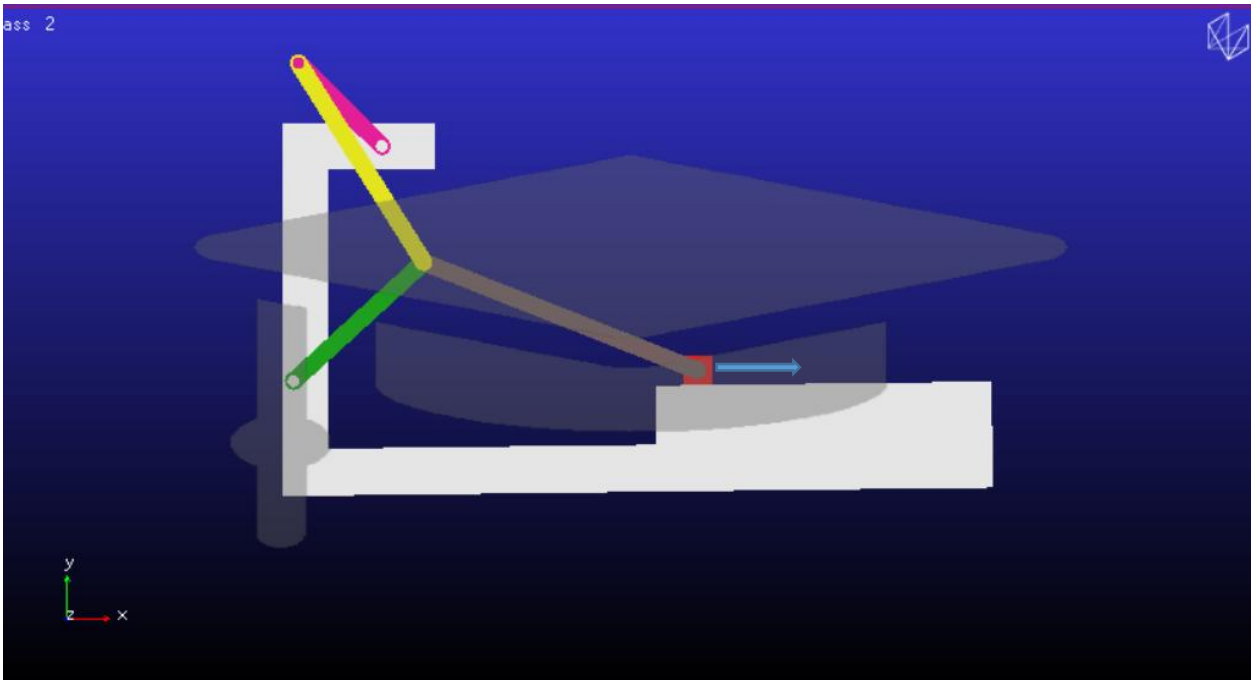


Fig 4. Mechanism at 135° position

Static force on 135°

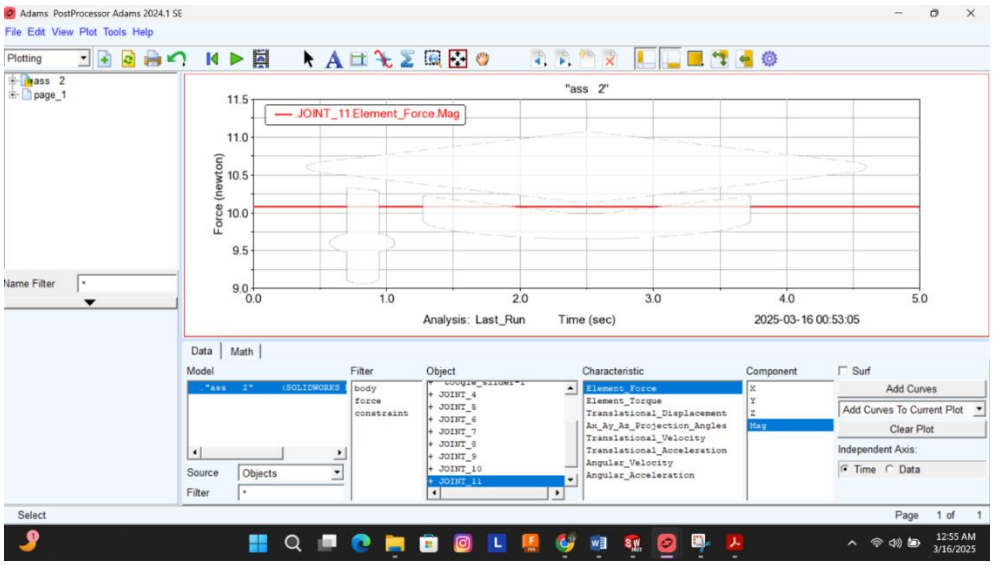


Fig 5. Force vs. Time

Static Torque on 135°

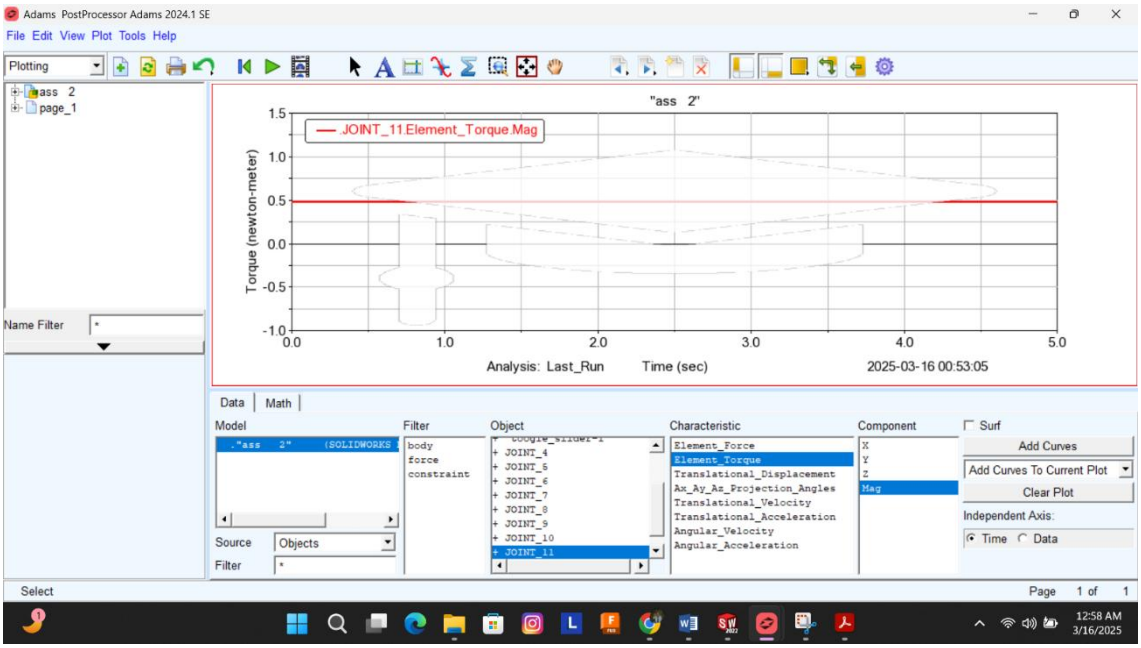


Fig 6. Torque vs. Time

From above figure it is seen that force 10.2N and torque 0.5 Nm in the crank is required to hold the mechanism in this equilibrium position.

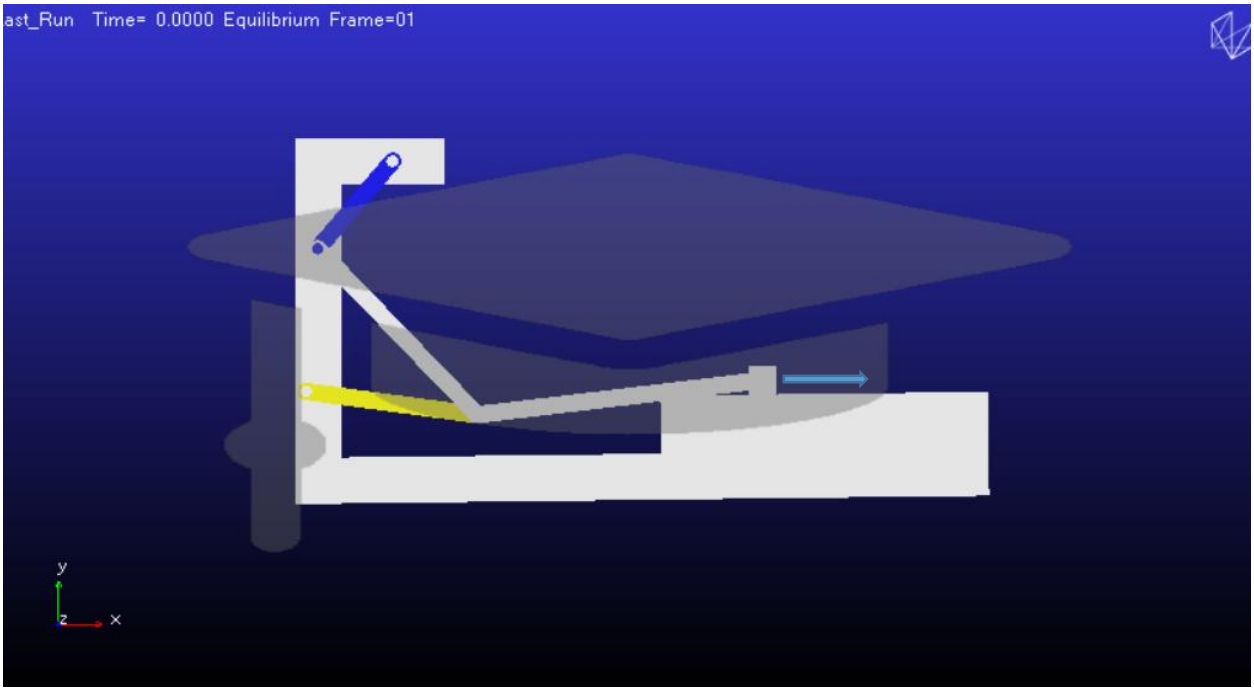


Fig 7. Mechanism at 225° position

Static force on 225°

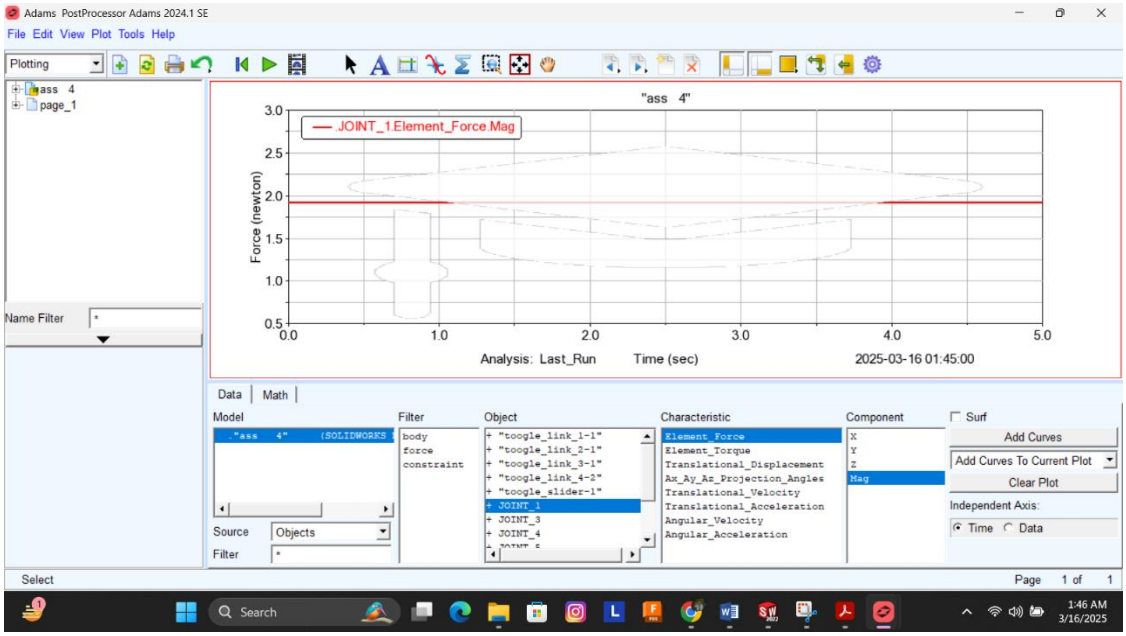


Fig 8. Force vs. Time

Static Torque on 225°

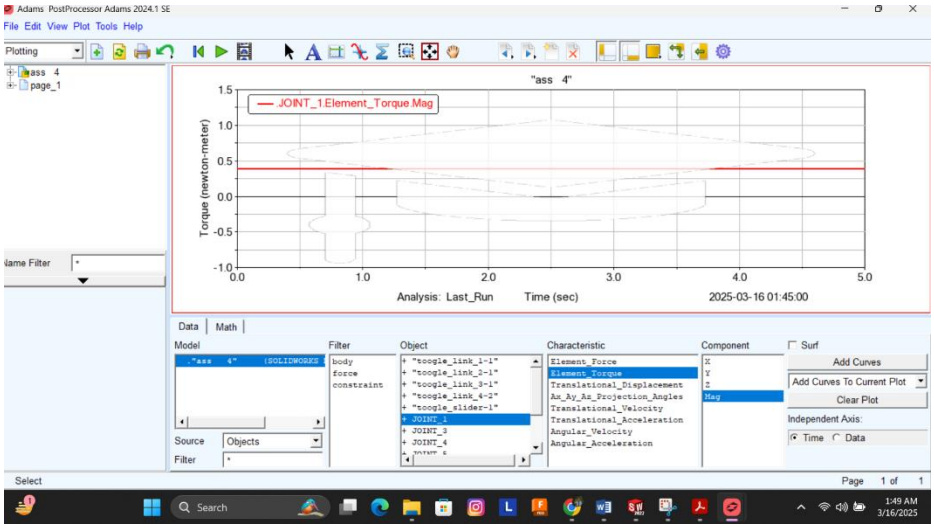


Fig 9. Torque vs. Time

From above figure it is seen that force 1.9N and torque 0.4 Nm in the crank is required to hold the mechanism in this equilibrium position.

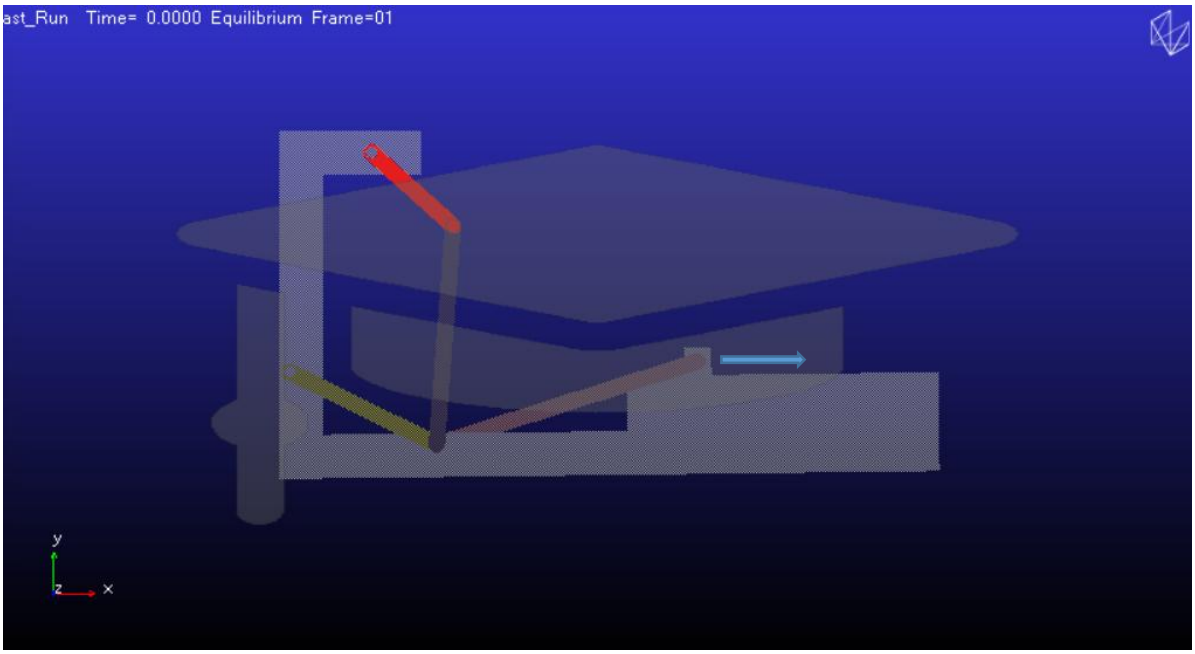


Fig 10. Mechanism at 315° position

Static force on 315°

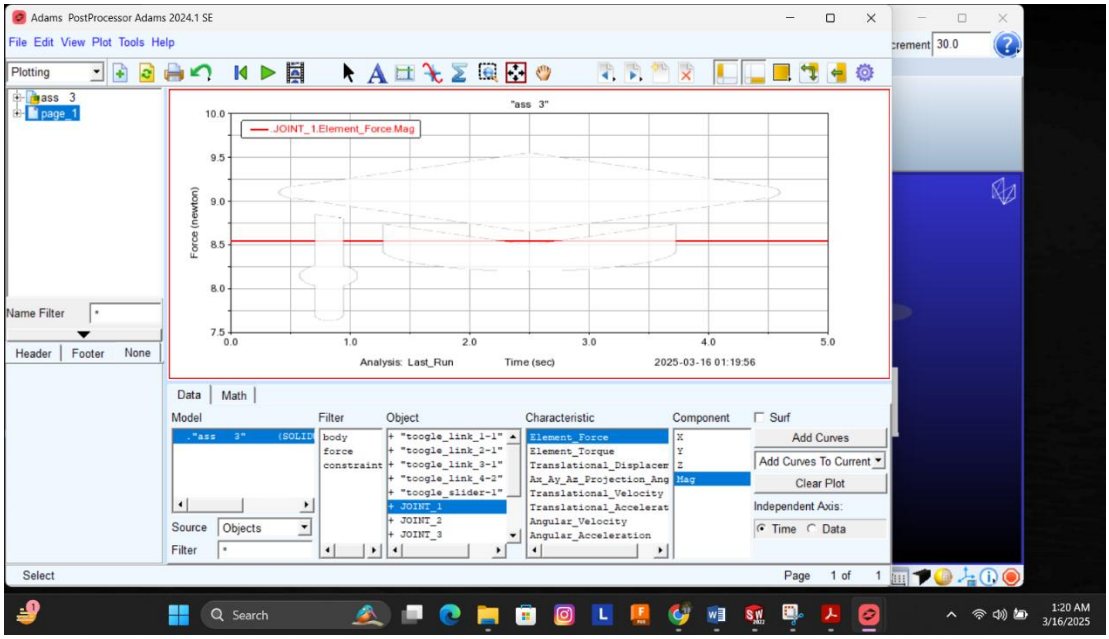


Fig 11. Force vs. Time

Static Torque on 315°

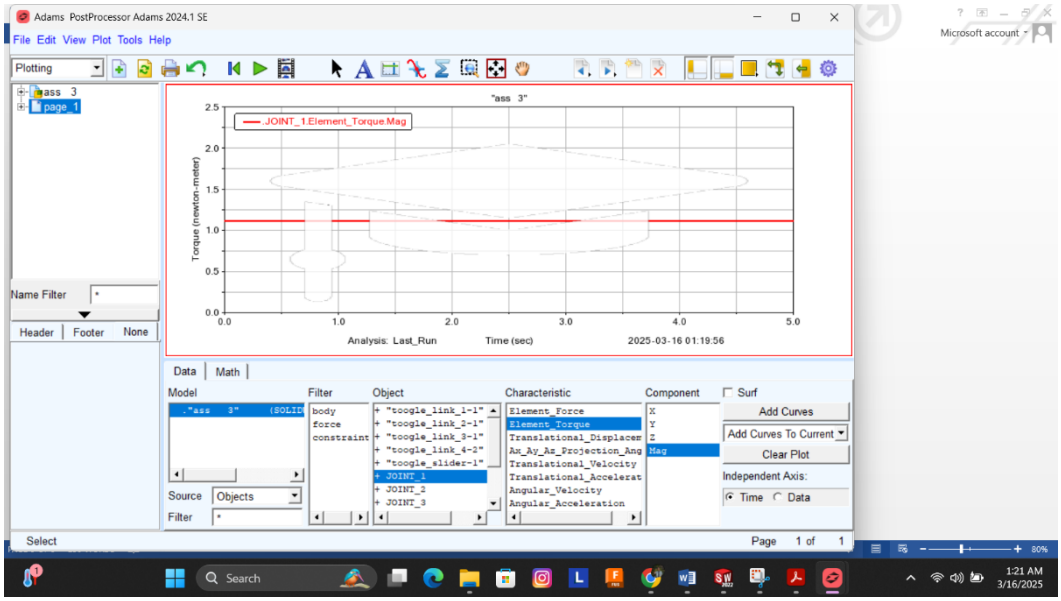


Fig 12. Torque vs. Time

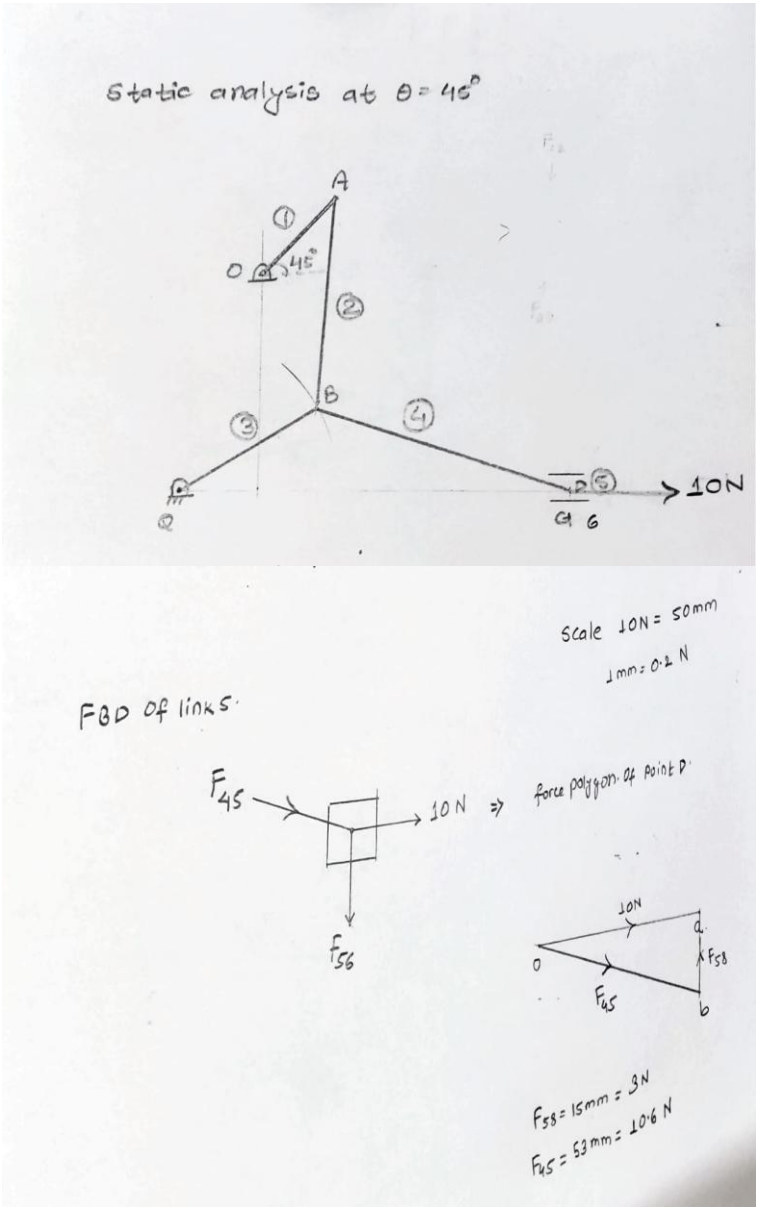
From above figure it is seen that force 8.55N and torque 1.14 Nm in the crank is required to hold the mechanism in this equilibrium position.

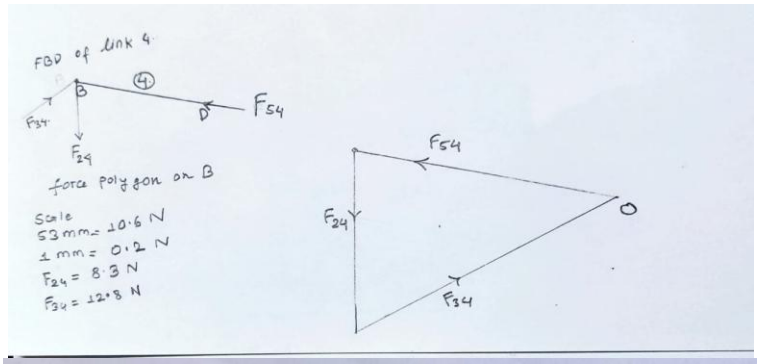
SECTION 2

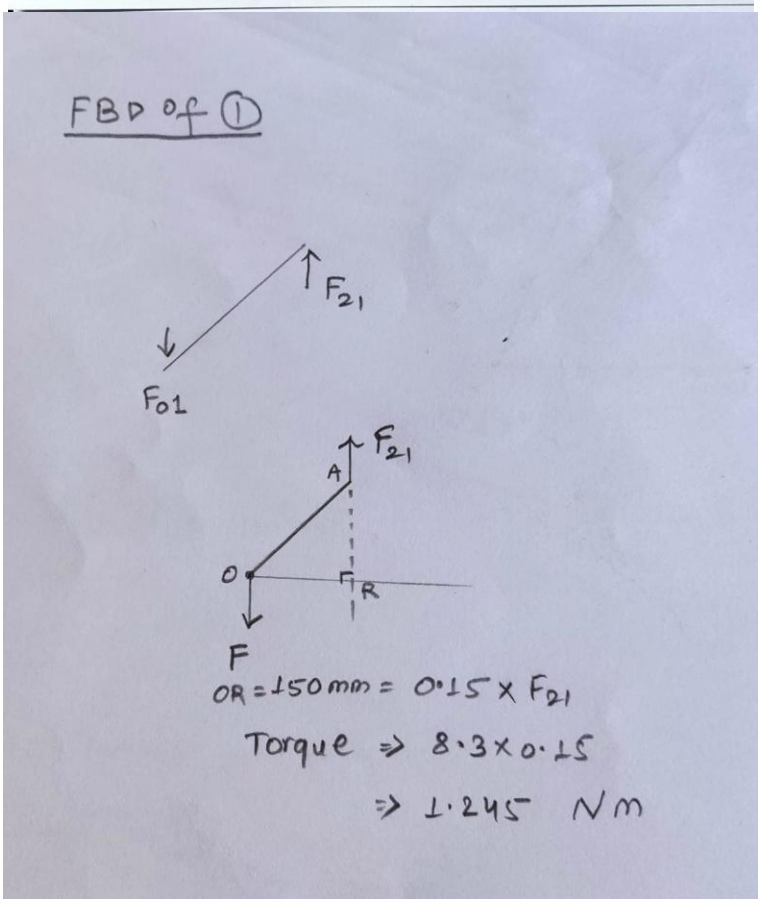
Validation of Static force analysis

I have done validation of static force using force polygon method which is easier Graphical method which is based on the principal of equilibrium of mechanism which states if mechanism is in equilibrium state then, all the linkage that forming the mechanism are also in equilibrium.

I have done validation at position when (Θ=45°)







From the above calculation the force and Torque required to hold the position in to equilibrium found to be 8.3N and 1.25Nm respectively which is close to simulated result. Some error may be due to step size taken for the simulation.

SECTION 3

Dynamic force analysis

In this section the dynamic analysis of the Variable 6 bar Toggle mechanism Is done by varying the torque applied at the crank. The resulting velocity And acceleration of the slider are plotted below.

Section 3.1: Constant driving torque

The driving component experiences a constant torque of 6 Nm, with its effects on the velocity and acceleration of the output component analyzed over a two cycle period. The applied torque significantly increases velocity amplitude, ranging from 15% to 30% between the first and second cycles. Similarly, the acceleration plot for the slider reveals a steady rise in acceleration values across Consecutive cycles. These observations indicate that the applied torque Significantly influences the velocity and acceleration profiles of the output Component, highlighting dynamic variations over the examined cycles.

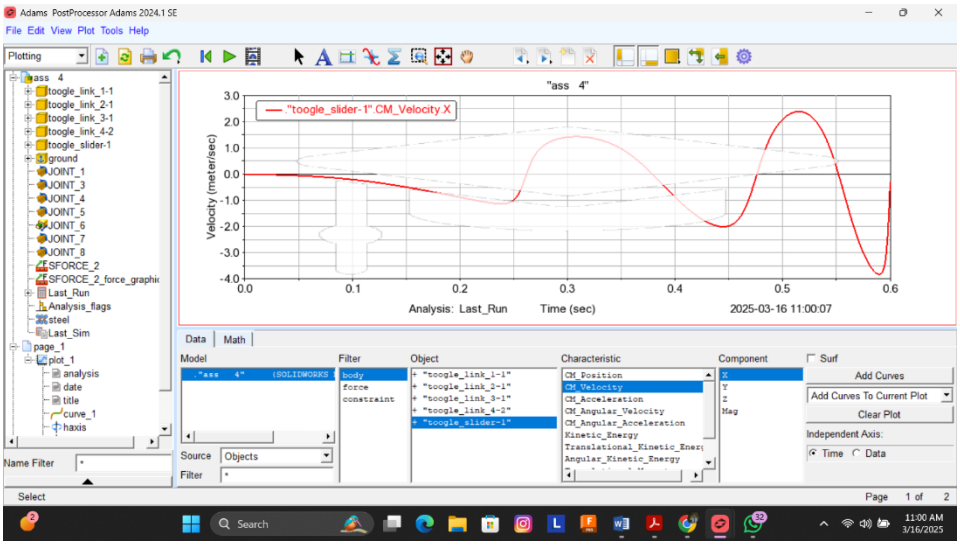


Fig 13. Velocity profile for Torque 6 Nm

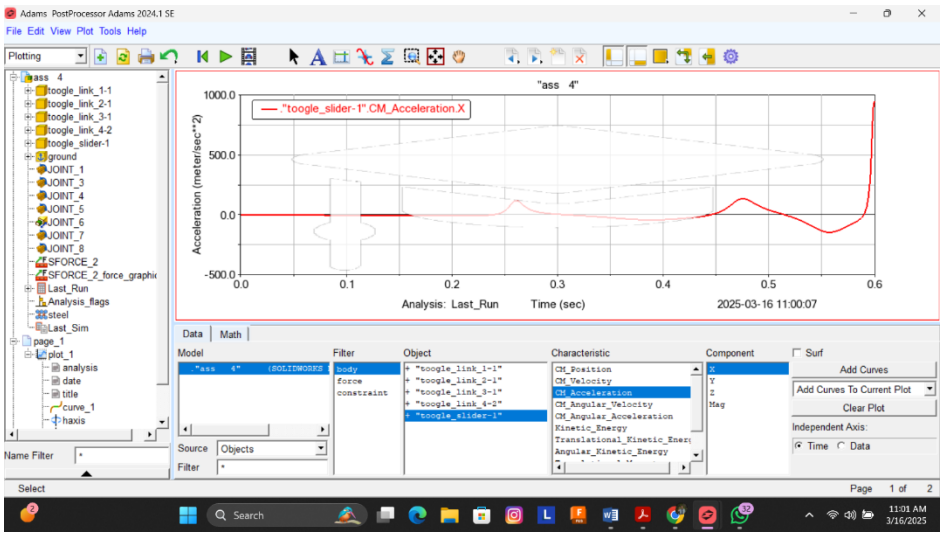


Fig 14. Acceleration profile for Torque 6 Nm

Section 3.2: Double the value of torque

In this section I have doubled the torque to 12 Nm from 6 Nm in which red curve shows profile for 6 Nm Torque and blue curve shows profile for 12 Nm Torque in the given figures.

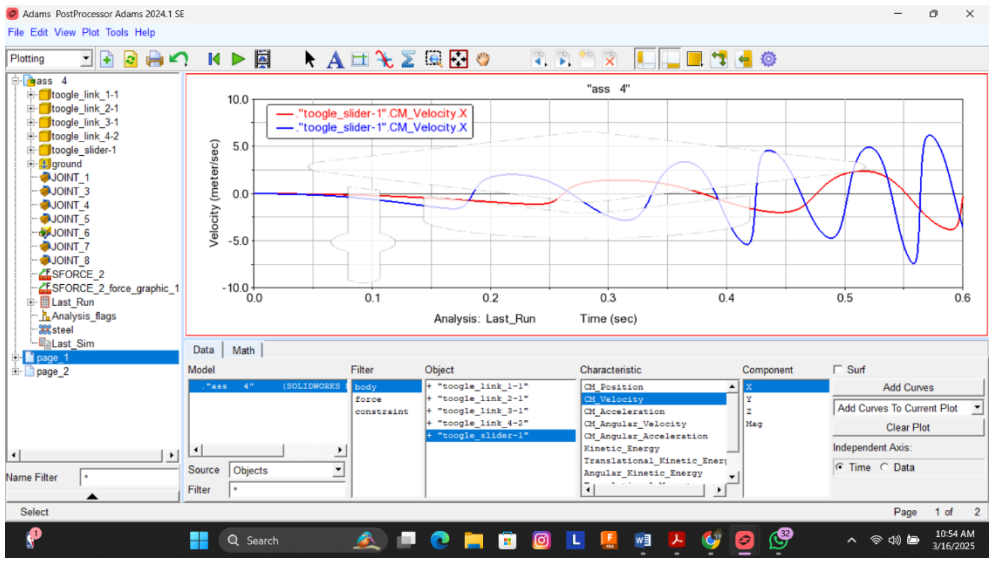


Fig 15. Velocity profile for Torque 6 Nm and 12 Nm

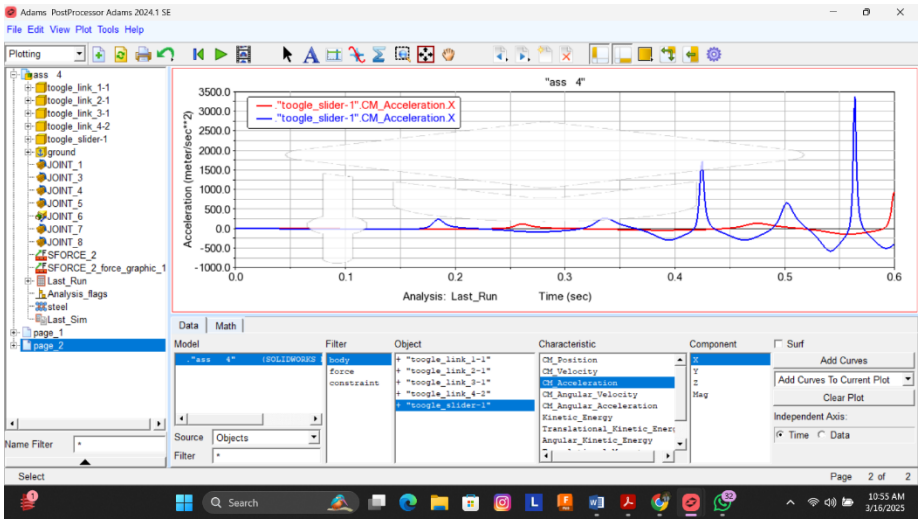


Fig 16. Acceleration profile for Torque 6 Nm and 12 Nm

Section 3.3: Sinusoidal driving torque

This section repeats the work done on section 3.1, with the torque being sinusoidal. The function of torque I have taken;

T(t) = 6*sin(21*t)

Here, T is torque, and $T = 6\ Nm$ with $\omega = 21\ rad/s$.

The Below shown graphs are the Velocity and acceleration profile when applied sinusoidal torque.

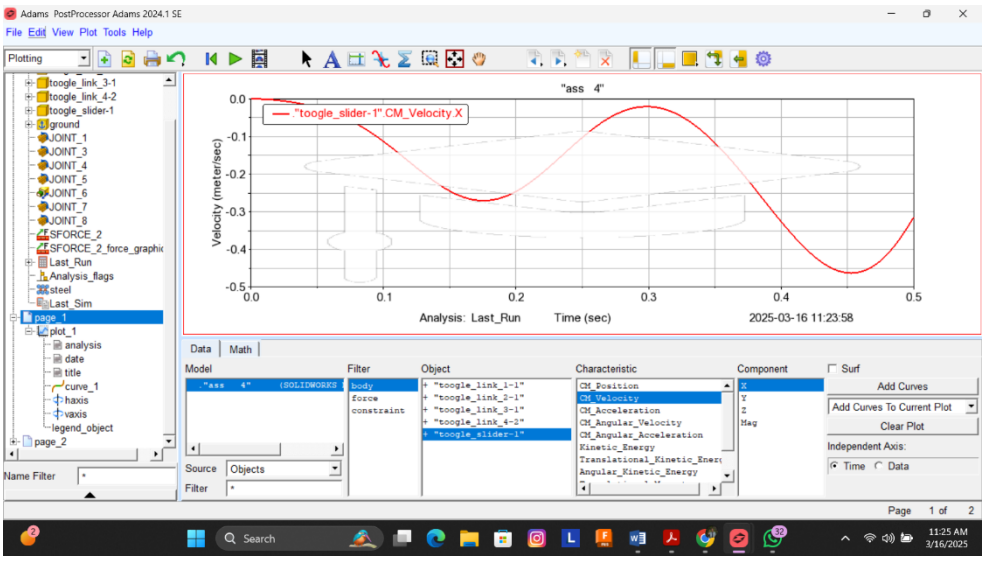


Fig 17. Velocity profile for sinusoidal Torque

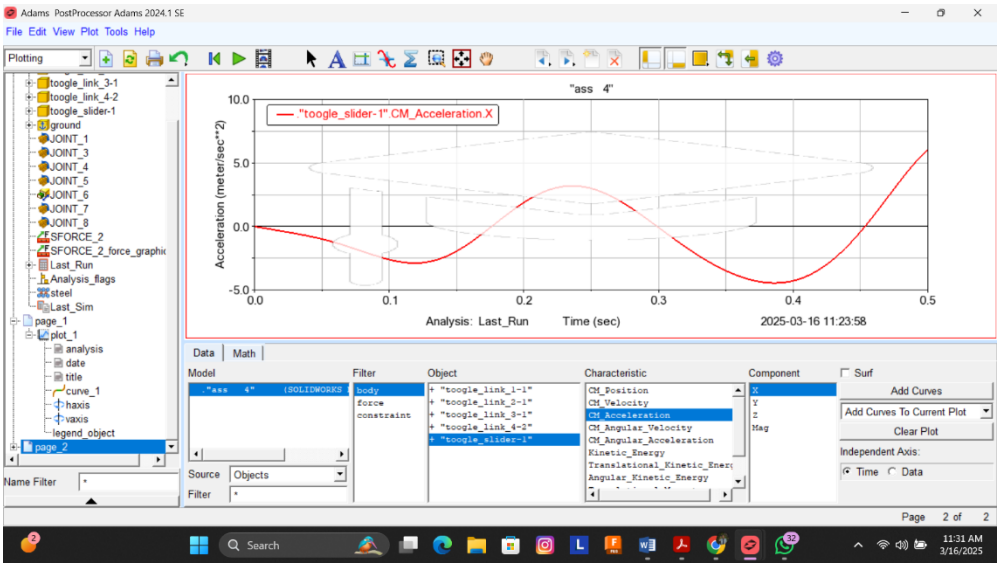


Fig 18. Acceleration profile for sinusoidal Torque

References:

- ***https://youtu.be/NgPeUdOKeVg?si=NnNNJSuAAK1X46MU***